KT24603 DATABASE

SEMESTER II, 2023/2024

LECTURER: DR MOHAMMED AHMED SALEH

ASSIGNMENT 1

MARATHON TRAINING SYSTEM

PREPARED BY:

BIL	NAME	MATRIC NUM
1	SYAZA SYAZANA BINTI ANWAR	[redacted]
2	SHWEHTTA PARANI KUMAR	
3	NORSYLAH BINTI SAFRUDIN	
4	WAN NURADRIANA BALQIS BT WAN AZMI	
5	LEENA A/P SELLAMANY	

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1.0 Introduction

Marathon training systems, accessible across different stages, have become indispensable tools for athletes preparing for long-distance races. These advanced companions offer a plethora of features meticulously designed to support runners at every stage of their training journey. From personalized training programs to advanced tracking interfaces, these systems serve as virtual coaches, nutritionists, and community hubs, revolutionizing the way competitors prepare for marathons.

At the heart of these systems lies the provision of personalized training programs. Leveraging advanced algorithms and expert knowledge, these programs are tailored to suit individual fitness levels, goals, and schedules. Whether aiming for a personal best or conquering a maiden marathon, users can access pre-designed workout regimens that provide structure and guidance throughout their training journey. This personalized approach ensures that competitors receive tailored training plans optimized for their specific needs and goals.

In addition to personalized training programs, marathon training systems often integrate tracking interfaces that allow users to monitor key performance metrics during their training sessions. Through the use of GPS technology, runners can track their routes, analyze elevation profiles, and record essential metrics such as pace, distance, and duration. These tracking features not only provide valuable insights into performance but also serve as motivational tools, allowing competitors to track their progress and celebrate their achievements along the way.

Furthermore, many marathon training systems offer a wealth of resources to support runners in areas beyond physical training. Nutrition and hydration play vital roles in marathon preparation, and these systems often provide guidance on fueling strategies and hydration plans to optimize performance and recovery. Users can access dietary tips, meal plans, and hydration reminders to ensure they are adequately fueled for their training sessions and races.

Additionally, injury prevention is a key focus of marathon training systems, with many incorporating modules dedicated to identifying and mitigating common risks. Through biomechanical analysis and expert insights, these systems can identify potential risk factors such as muscular imbalances, stride abnormalities, and overtraining patterns. Users receive personalized recommendations and corrective exercises to address these issues, reducing the likelihood of injuries and promoting long-term athletic resilience.

In conclusion, marathon training systems provide a comprehensive solution for athletes preparing for long-distance races. With features ranging from personalized training programs to advanced tracking interfaces and injury prevention modules, these systems cater to the diverse needs of runners at every level of experience and expertise. By leveraging the capabilities of these systems, competitors can optimize their training experience and achieve their marathon goals with confidence and determination.

1.1 Users and their Roles

Within the realm of marathon training apps, runners constitute the core user base, encompassing a broad spectrum of individuals with diverse backgrounds and objectives. These users engage with the app as a pivotal tool in their marathon preparation, leveraging its features to optimize training, monitor progress, and connect with fellow runners.

Amateur runners, often new to long-distance running, seek guidance and support tailored to their level of experience. They benefit from beginner-friendly training plans, instructional content, and motivational resources aimed at building endurance and confidence. In contrast, experienced runners, boasting prior marathon experience, gravitate towards advanced training plans and detailed performance analytics to refine their strategies and achieve specific goals.

For professional athletes, the app serves as a sophisticated training tool, offering specialized plans, advanced analytics, and integration with cutting-edge equipment to maximize their competitive edge. Recreational runners, who run for enjoyment and fitness, value flexibility in training plans and community features that foster motivation and camaraderie.

Goal-oriented runners, driven by specific objectives, rely on the app's goal-setting and progress-tracking features to monitor their achievements and stay focused on their targets.

1.2 Problem with Existing System

With the increasing number of users, it is crucial to ensure a seamless user experience on a robust database management system. Systems such as the Marathon Training system are in high demand where there is significant rise in the number of users. Based on the modules of our Marathon Training System which includes Training tracking, Health tracking and User authentication, it may be capable of leading to several system problems such as scalability challenges and data accuracy.

As per the User authentication, scalability challenges may occur in such due to the growth in the user base. The system may struggle to scale effectively to accommodate increased data volume and user activity. The organization that manages the app would not have their database configured to expand efficiently. There are certain factors where the app may not be designed to support. For example, when the data is updated and queried, the queries become more complex. Most relational databases are not designed to support such things. As a result the app may experience several issues such as performance degradation and downtime during peak usage hours.

Besides, in terms of the Health Tracking module, issues like data privacy and security may arise.
It may be risky with the storing and transmitting of sensitive health information. Moreover,
It has the potential for data breaches leading to unauthorized access to users' health data. It fails to maintain users' privacy and trust towards the application. Lack of transparency in data handling practices makes it hard in the implementation of robust encryption and access control.
Not only that, the module may also face challenges like the integration challenges. For instance, difficulties may occur in Application Programming Interface (API) which is offered by the health monitoring applications to integrate the data.

Apart from that, for modules such as the training tracking, concurrency issues may occur in the marathon training system when many users access and modify the same data at the same time. This issue will result in data inconsistency and integrity problems within the database and also lost updates in the application. This may lead to users losing their training records or encounter inaccurate data. Concurrency control is a must in a DBMS to ensure the data integrity is preserved.

Furthermore, systems such as the Marathon Training system may suffer from slow performance for certain aspects. If the database is not optimized for efficient data retrieval and storage, the app is more likely to face slow performance especially when accessing or updating user information. Sluggish response time during data retrieval or synchronization processes can frustrate the users as a result of slow performance. This may also hinder their training experience as it takes too long to respond or when loading.

In summary, there are certain problems that a system may address on the database aspect. It is crucial to provide a better performing system for the users so that the data stored is secured and updated from time to time. The Marathon Training system can be improvised on the listen modules and others as well if methods taken to make it more efficient for the users to use.

2.0 Business Rules

Business Rules are anything that explains what the system is about. In a database, it defines the policies, processes, and limitations that govern data management and use within an organization or system. In the context of a marathon training system, business rules might include principles for handling training plans, participant data, and event planning. Below are lists of business rules of the Marathon Training system that is designed.

- A user can register one account. Each account is registered by a single user.
- A user can participate in many events. An event is participated by one or more users.
- A user can follow one or more training plans. A training plan is followed by many users.
- A user can have one or more medical records. Each medical record is owned by a single user.
- A user can have many discussion groups. Each discussion group is generated by a single user.
- An event can include one or more training plans. Each training plan is included in one or more events.
- An event can have one or more medical records. Each medical record is accessible by many events.
- An event can create one or more discussion groups. Each discussion group is created for a single event.
- A training plan can access one or more medical records. Each medical record is accessible by many training plans.
- A training plan can have many discussion groups. Each discussion group is created for a single training plan.

3.0 Entity-Relationship Model (ERM) components

<u>Entities</u>

1. User
2. Event
3. Training Plan
4. Medical Records
5. Discussion Group

<u>Attributes</u>

1. User Authentication

- Username
- Password (encrypted)
- Email
- Security question(s)
- Security answer(s)

2. Event Management

- Event ID
- Event name
- Event date/time
- Location
- Description
- Organizer
- List Participants
- Registration deadline
- Event type
- Registration fee

3. Training Tracking Plan

- Plan ID

- Plan name
- Duration
- Start date
- End date
- Weekly training schedule
- Mileage targets
- Training intensity levels
- Progress tracking

4. Medical Records

- Record ID
- User ID (foreign key linking to User Authentication)
- Date of visit
- Medical condition(s) include allergies
- Medications
- Treatment history include (Previous surgeries etc)

5. Discussion Group

- Group ID
- Group name
- Description
- Members
- Topics
- Posts
- Moderator(s)
- Date created
- Last activity date/time

<u>Relationship</u>

1. Event Management Module
    - Event↔User : Event include users as participant.

- Event↔Training Plan : Events associated with training plans.
- Event↔Medical Records : Events need medical records.
- Event↔Discussion Groups : Event creates discussion group.

2. Training Tracking Module
    - Training Plan↔User : Training plan include users.
    - Training Plan↔Event : Training plans associated with events.
    - Training Plan↔Medical Records : Training plans need medical records.
    - Training Plan↔Discussion Groups : Training plan created discussion groups.
3. Health Tracking Module
    - Medical Records↔User : Medical records are generated by users.
    - Medical Records↔Training Plan : Medical records affecting training plans.
    - Medical Records↔Event : Medical records is given to events.
4. User Authentication Module
    - User↔User : User authenticate with the system.
    - User↔Event : User participate in events.
    - User↔Training Plan : User follow training plans.
    - User↔Medical Records : User generated medical records.
    - User↔Discussion Groups : User joins discussion group.
5. Community Module
    - Discussion Groups↔User : Discussion groups are joined by users.
    - Discussion Groups↔Event : Discussion groups is created for events.
    - Discussion Groups↔Training Plan : Discussion groups is created for training plans.

<u>Constraints</u>

User:

- user_id: Primary Key
- username: Not Null, Unique
- password: Not Null
- email: Not Null, Unique
- security_question: Not Null

- security_answer: Not Null
- last_login_date_time: Null able
- account_creation_date_time: Not Null

Event:

- event_id: Primary Key
- event_name: Not Null
- event_date_time: Not Null
- location: Not Null
- description: Null able
- organizer: Not Null
- registration_deadline: Not Null
- event_type: Not Null
- registration_fee: Null able
- organizer_id: Foreign Key referencing User (user_id)

TrainingPlan:

- plan_id: Primary Key
- plan_name: Not Null
- duration: Not Null
- start_date: Not Null
- end_date: Not Null
- weekly_training_schedule: Not Null
- mileage_targets: Not Null
- training_intensity_levels: Not Null
- progress_tracking: Not Null

MedicalRecord:

- record_id: Primary Key
- user_id: Foreign Key referencing User (user_id), Not Null
- date_of_visit: Not Null
- medical_conditions: Null able

- medications: Null able
- allergies: Null able
- previous_surgeries: Null able
- test_results: Nullable
- treatment_history: Nullable

DiscussionGroup:

- group_id: Primary Key
- group_name: Not Null
- description: Nullable
- topics: Nullable
- posts: Nullable
- date_created: Not Null
- last_activity_date_time: Nullable
- moderator_id: Foreign Key referencing User (user_id)

UserDiscussionGroup:

- user_id: Foreign Key referencing User (user_id), Not Null
- group_id: Foreign Key referencing DiscussionGroup (group_id), Not Null

Dependencies

1. Mandatory Dependencies
    - Event↔Medical Records : Events must need participants' medical records.
    - Event↔User : Event must include users as participant.
    - Event↔Training Plan : Events must be associated with training plans.
    - Event↔Discussion Groups : Events must create discussion groups.
    - Training Plan↔User : Training plan must include users.
    - Training Plan↔Medical Records : Medical record must be linked to training plan.
    - Medical Records↔User : Medical records must be generated by users.
    - Medical Records↔Event : Medical records must be given to events.

- User↔User : Users must authenticate with the system.
- User↔Training Plan : Users must follow training plans.
- User↔Medical Records : Users must have medical records.
- Discussion Groups↔User : Discussion groups must be joined by users.

2. Optional Dependencies:
   - User↔Event : Users may or may not participate in events.
   - Training Plan↔Event : Training plans may or may not be associated with events.
   - Training Plan↔Discussion Groups : Training plan may or may not created discussion groups.
   - Medical Records↔Training Plan : Medical records may or may not affecting training plans.
   - User↔Event : User may or may not participate in events.
   - User↔Discussion Groups : Users may or may not participate in discussion groups.
   - Discussion Groups↔Event : Discussion groups may or may not be created for events.
   - Discussion Groups↔Training Plan : Discussion groups may or may not be created for training plans.

4.0 Modules in the System

1. User Authentication Module

This module manages user accounts, authentication and the security features within the application. Users can register for new accounts by providing their information like usernames,emails,passwords and also security details. Once registered,users can log in using their username and password. For the recovery purpose,users can find their existence account by answering security questions that have been registered before. Account creation and the login functionalities are available to all the users. Additionally, this module also includes features of updating account information and password. CRUD operations include creating new accounts,reading account information,updating user account details and deleting or deactivating accounts.

2. Training Tracking Plan Module

This module is designed to assist users in planning and monitoring their progress. This module includes tools for creating ,managing and tracking their performance by their personalized training plans based on the specific objectives and preferences. Users can create the training plan based on the suggestion given by entering pertinent details such as the plan's name,target race distance or mileage ,marathon or half marathon and the duration of the marathon. Once created the training plan,users are able to access and track their progress. Additionally, CRUD also integrated into this module’s functionality. Users can create new training plans based on the details of the marathon,like the name of the marathon,duration and dates to plan their specific needs and objectives. As to read users can access and view their existing training plan and can arrange their time wisely.(Update) users can modify their training plan as needed,users can adjust the parameters such as the duration, milage target ,training intensity levels and weekly schedules to accommodate changes based on the marathon event.(Delete) users have the option to delete the training plan once the event is finished and no longer relevant, or needed.

3. Health Tracking Module

This module acts as a central hub for users to manage their medical data. Through seamless integration of CRUD, users can effectively document, review, update, and remove medical records as needed. This is to make sure users keep an accurate and current health profile within the app. By entering relevant information like visit dates, registered medical conditions including allergies, prescription medications, and thorough treatment histories including previous surgeries within the module, users may easily build new medical records. This function guarantees that users may effectively record and capture crucial health information, enabling prompt access and retrieval when needed. Moreover users can keep aware about their health state,track changes over time and integrate it with the training tracking plan that considers their health and minimize their injury risks. As for the CRUD operations, (create) users can create new medical records by inputting relevant information such as the date of visit, medical conditions, medication and treatment history. (read) users can access and review their existing medical records including the details on it which enable users to stay informed with their health status. (Update) users have the ability to update the medical records as necessary. Users can change it based on their up to date health condition. (delete) Users have the access to remove the unnecessary medical records to organize the medical record with up to date information of the user's health to ensure only relevant information is retained.

4. Community Module

The Community module of Marathon Training System is equipped with a database to support group dialogues by means of forums, chat rooms and virtual coaching support. The database schema has the "CommunityPosts" table, in which forum posts, chat messages, and coaching comments are stored. Fields such as post ID, user ID for authorship attribution, content, timestamp and associated metadata are used. This database-driven approach provides a smooth and seamless CRUD (Create, Read, Update, Delete) operation that enables users to participate in the app's community ecosystem. Users can craft new forum threads, make posts, or read coaching advice, all of which are then stored in the "CommunityPosts" table and personalized data. Those abilities provide users with a chance to pull community content, filtering by the thread, user ID, or time stamp, which in turn makes getting required information to be fast and easy. In addition, the users have the privilege to update their posts or messages, with the modifications appearing in the database, and they can also delete their own content, keeping the control of their contributions. This method of structuring the data in the database and CRUD operations allows users to quickly and easily interact with the community which keeps the Marathon Training system highly active.

5. Event Management

The Event Management Module of the Marathon Training System where all the information about the marathons events to be held in the future is stored using a database function which manages the details. The database structure consists of the "Events" table which contains the event ID, name, date, location, and registration status. This database-driven approach leads to the ability to carry out CRUD operations and the effective management of event data via DBMS (Database Management Systems). When a new marathon event is added, a new item is connected in the "Events" table, where all the needed information is stored. Next, users can use the "Events" table to get information about upcoming events, filtering the results based on criteria like date or location, so that they can keep up with the latest event offers. In a scenario involving any changes to event details like rescheduling or venue updates, the relevant entries in the "Events" table are updated correspondingly. On the other hand, an event to be canceled or completed, its record is then removed from the "Events" table to ensure that the database remains precise and current. This database structure and the associated CRUD operations are the basis on which the information about the marathon event can be managed effectively, thereby ensuring that users have access to the latest and most relevant details in the app.

5.0 Identification of Primary Key, Foreign Key and Composite Key

Primary Keys

1. User
    - ‘user_id’
2. Event
    - ‘event_id’
3. Training Plan
    - ‘plan_id’
4. Medical Record
    - ‘record_id’
5. Discussion Group
    - ‘group_id’

Foreign Keys

1. Event
    - ‘organizer_id’
2. Training Plan
    - ‘user_id’
3. Medical Record
    - ‘user_id’
4. Discussion Group
    - ‘moderator_id’
5. Event Management
    - ‘event_id’
    - ‘organizer_id’

Composite Keys

1. User Discussion Group
    - ‘user_id’
    - ‘group_id’

6.0 Normalization Process (1NF-3NF)

USER AUTHENTICATION

Before:

UserID	Username	Password	Email	SecurityQuestions	SecurityAnswers
U001	user1	pass1	email1@example.com	What is your pet's name?; What is your birthplace?	Fluffy; City X
U002	user2	pass2	email2@example.com	What is your mother's maiden name?	Smith

1NF :

UserID	Username	Password	Email	SecurityQuestions	SecurityAnswers
U001	user1	pass1	email1@example.com	What is your pet's name?	City X
U001	user1	pass1	email1@example.com	What is your birthplace?	Fluffy
U002	user2	pass2	email2@example.com	What is your mother's maiden name?	Smith

2NF:

Username	Password (encrypted)	Email
user1	abc123!@#	user1@example.com
user2	def456$%^	user2@example.com

Username	Security question	Security answer
user1	What is your pet's name?	Fluffy
user1	What is your mother's maiden name?	Smith
user2	What was your first car?	Toyota
user2	What is your favorite book?	1984

3NF:

Already in 3NF, as there are no transitive dependencies.

EVENT MANAGEMENT

Before:

Event ID	Event name	Event date/time	Location	Description	Organizer	List Participants	Registration deadline	Event type	Registration fee
1	Marathon	2024-06-01 08:00	City Park	Annual marathon	org1	user1, user2	2024-05-15	Sport	50
2	Running	2024-07-10 10:00	Park	Morning running session	org2	user1	2024-07-01	Wellness	20

1NF:

Event ID	Event name	Event date/time	Location	Description	Organizer	Participant	Registration deadline	Event type	Registration fee
1	Marathon 2024	2024-06-15 08:00	City Park	Annual marathon event	org1	user1	2024-06-01	Marathon	50
1	Marathon 2024	2024-06-15 08:00	City Park	Annual marathon event	org1	user2	2024-06-01	Marathon	50
2	Half Marathon	2024-07-20 07:00	Riverside	Half marathon run	org2	user1	2024-07-05	Half-marathon	30

2NF:

Event ID	Event name	Event date/time	Location	Description	Organizer	Registration deadline	Event type	Registration fee
1	Marathon	2024-06-01 08:00	City Park	Annual marathon	org1	2024-05-15	Sport	50
2	Running	2024-07-10 10:00	Park	Morning running session	org2	2024-07-01	Wellness	20

Event ID	Participant Username
1	user1
1	user2
2	user1

3NF:

The structure remains the same as there are no transitive dependencies in the 2NF tables.

TRAINING TRACKING PLAN

Before:

Plan ID	Plan name	Duration	Start date	End date	Weekly training schedule	Mileage targets	Training intensity levels	Progress tracking
1	Marathon Prep	16 weeks	2024-01-01	2024-04-21	Mon: 5 miles, Wed: 8 miles, Fri: 10 miles	500 miles	High	On track
2	Marathon for Beginners	8 weeks	2024-05-01	2024-06-25	Mon:3 miles, Wed: 4 miles, Fri: 5 miles	250 miles	Moderate	Progressing

1NF:

Plan ID	Plan name	Duration	Start date	End date	Mileage targets	Training intensity levels	Progress tracking
1	Marathon Prep	16 weeks	2024-01-01	2024-04-21	500 miles	High	On track
2	Marathon for Beginners	8 weeks	2024-05-01	2024-06-25	N/A	Moderate	Progressing

Plan ID	Day	Activity
1	Monday	5 miles run
1	Wednesday	8 miles run
1	Friday	10 miles run
2	Monday	4 miles run
2	Wednesday	3 miles run
2	Friday	5 miles run

2NF:

Already in 2NF.

3NF:

Already in 3NF, as there are no transitive dependencies.

MEDICAL RECORDS

Before:

Record ID	User ID	Date of visit	Medical condition(s)	Medications	Treatment history
1	user1	2024-02-10	Asthma,allergies	Inhaler	Annual check-ups, No surgeries
2	user2	2024-03-15	Allergies,asthma	Antihistamines	Allergy shots

1NF:

Record ID	User ID	Date of visit	Medical condition	Medication	Treatment history
1	user1	2024-02-10	Asthma	Inhaler	Annual check-ups, No surgeries
1	user1	2024-02-10	Allergies	Antihistamines	Annual check-ups, No surgeries
2	user2	2024-03-15	Allergies	Antihistamines	Allergy shots, Surgery on knee
2	user2	2024-03-15	Diabetes	Insulin	Allergy shots, Surgery on knee

2NF:

In the current table, Record ID is the primary key, and all attributes are fully dependent on Record ID.No partial dependencies are present, so the table is already in 2NF.

3NF:

MedicalRecords:

Record ID	User ID	Date of visit	Medical condition	Medication
1	user1	2024-02-10	Asthma	Inhaler
1	user1	2024-02-10	Allergies	Antihistamines
2	user2	2024-03-15	Allergies	Antihistamines
2	user2	2024-03-15	Diabetes	Insulin

TreatmentHistory:

Record ID	Treatment history
1	Annual check-ups, No surgeries
2	Allergy shots, Surgery on knee

DISCUSSION GROUP

Before:

Group ID	Group name	Description	Members	Topics	Posts	Moderator(s)	Date created	Last activity date/time
1	Marathon Runners	For marathon enthusiasts	user1, user2	Training Tips, Race Day Prep	Stretch before running!, Hydrate well!	user1	2024-01-15	2024-05-01
2	Marathon Beginners	For new marathon runners	user1	Beginner Poses, Breathing Techniques	Breath deep and slow	user2	2024-03-10	2024-05-05

1NF:

DiscussionGroup:

Group ID	Group name	Description	Date created	Last activity date/time
1	Marathon Runners	For marathon enthusiasts	2024-01-15	2024-05-01
2	Marathon Beginners	For new marathon runners	2024-03-10	2024-05-05

GroupMembers:

Group ID	Member Username
1	user1
1	user2
2	user1

GroupTopic:

Group ID	Topic ID	Topic name
1	1	Training Tips
1	2	Race Day Prep
2	3	Beginner Poses
2	4	Breathing Techniques

GroupPosts:

Post ID	Group ID	Topic ID	Content	Post date
1	1	1	"Remember to stretch!"	2024-02-01
2	1	2	"Hydrate well!"	2024-04-20
3	2	3	"Mountain Pose tutorial"	2024-04-15
4	2	4	"Breath deep and slow"	2024-04-25

GroupModerators:

Group ID	Moderator Username
1	user1
2	user2

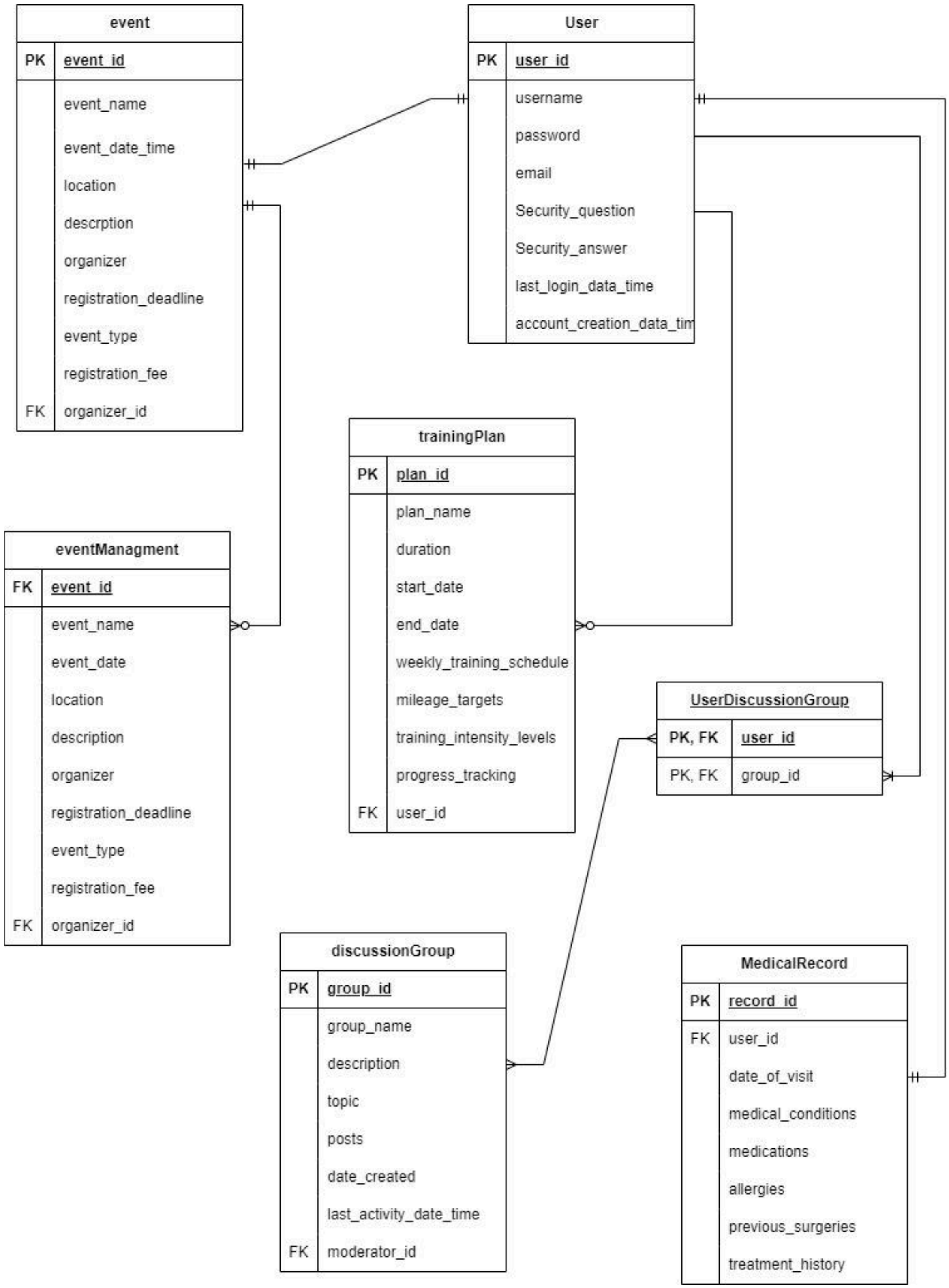
2NF:

In this case, there are no partial dependencies, so the structure remains the same.

3NF:

The structure already satisfies 3NF requirements since there are no attributes that depend on non-key attributes. Each table represents a unique entity with its attributes properly defined.The structure of entity 5 (Discussion Group) remains unchanged after 2NF and 3NF normalization, as it already conforms to both forms without requiring further modification.

7.0 Entity-Relationship Diagram (ERD) Model



8.0 Conclusion

In conclusion, the database design of the Marathon Training System serves as the backbone, ensuring effective data storage, retrieval, and management. The specified business rules provide a solid foundation for designing the database schema, allowing for smooth interaction between users and the system. From user authentication to event management and health tracking, each module is methodically built to meet specific user requirements while maintaining data integrity and security.

However, as the system matures and user demand increases, it is critical to stay cautious in addressing scalability issues and data privacy concerns. The system can handle increased data volume and provide uninterrupted service during peak usage hours by employing strong database management principles such as efficient indexing, normalization, and backup strategies.

Furthermore, ongoing monitoring and adjustments to the database structure will be required to accommodate changing user requirements and technology improvements. By remaining fluid and responsive to user feedback, the Marathon Training System can continue to improve as a dependable and indispensable tool for athletes preparing for long-distance events, allowing them to achieve their objectives successfully and securely.